

Robust Fine-Tuning of Large Language Models for Retrieval Augmented Generation in Power Grid Dispatching Regulations

Ying Zou *

Institute of Artificial Intelligence
China Electric Power Research
Institute Co., Ltd
Beijing, China
zouying@epri.sgcc.com.cn

Ying Liu

Institute of Artificial Intelligence
China Electric Power Research
Institute Co., Ltd
Beijing, China
liuying3@epri.sgcc.com.cn

Sai Zhang

Institute of Artificial Intelligence
China Electric Power Research
Institute Co., Ltd
Beijing, China
zhangsai@epri.sgcc.com.cn

Bochuan Song

Institute of Artificial Intelligence
China Electric Power Research
Institute Co., Ltd
Beijing, China
songbochuan@epri.sgcc.com.cn

Xiao Liang

Institute of Artificial Intelligence
China Electric Power Research
Institute Co., Ltd
Beijing, China
liangxiao1@epri.sgcc.com.cn

Abstract—This paper presents an retrieval-augmented based question-answering model tailored for power grid dispatch regulations. We introduce an automated high-quality fine-tuning dataset generation approach based on the parsed power grid dispatch regulations clauses for question-answering model. Additionally, a robust retrieval-augmented generation fine-tuning method using Low-Rank Adaptation is employed to strengthen the model's resilience against irrelevant information. Experimental results demonstrate the effectiveness of the proposed method, the model achieves improvements in accuracy compared to baseline models. This method, by integrating automated dataset generation, retrieval-augmented generation, and robust fine-tuning, offers a reliable and efficient solution for question-answering tasks in the power grid dispatch domain.

Keywords—Power grid dispatch, Large Language Model(LLM), Retrieval-Augmented Generation (RAG)

I. INTRODUCTION

In recent years, large language models (LLMs)[1] have made significant strides in natural language processing. These models have shown great potential in tasks such as question answering[2], document understanding[3], and domain-specific applications. However, the integration of LLMs into particular domains, particularly in technical domains like power grid scheduling, remains a challenging endeavor due to the complex nature of specialized knowledge and the need for precise information retrieval[4].

When applying large language models (LLMs) to specialized domains, Retrieval-Augmented Generation (RAG) [5] and Supervised Fine-Tuning (SFT) are two prominent approaches. RAG enhances the accuracy and timeliness of LLMs by integrating external knowledge bases with their generative capabilities, enabling dynamic retrieval of relevant information during question answering. However, the effectiveness of RAG is constrained by its reliance on the quality and coverage of external knowledge bases [6]. Furthermore, its heavy dependence on the accuracy of the

retrieval model may introduce noise during the retrieval process, thereby compromising the relevance and consistency

of the generated outputs. . On the other hand, Supervised Fine-Tuning adapts pre-trained models to specific tasks by fine-tuning them on domain-specific datasets. While this approach significantly improves model performance in specialized fields, it faces challenges due to its dependence on high-quality annotated data. Such data is often scarce and costly to obtain in specialized domains, which can lead to over-fitting or limited generalization capabilities. Additionally, SFT struggles to adapt to rapidly evolving domain knowledge, restricting its applicability in dynamic environments. Moreover, fine-tuned models are prone to generating hallucinated information when encountering unfamiliar or unseen knowledge [8][9], further highlighting the need for robust solutions to address these limitations.

This research aims to combine the strengths of both RAG and fine-tuning approaches by proposing a robust fine-tuning strategy, Robust Retrieval Augmented Generation Fine Tuning[10]. Robust Retrieval Augmented Generation Fine Tuning seeks to address the limitations of both methods, enhancing the model's ability to learn domain-specific patterns while simultaneously improving its robustness in handling irrelevant or noisy data during the retrieval and answer generation process. In this study, we focus on the application of Robust Retrieval Augmented Generation Fine Tuning to power grid dispatch regulations document clauses, a domain with highly structured and specialized knowledge. We first parse the power grid dispatch regulations documents and break them down into individual clauses, extracting key elements such as filename, clause title, clause number, and clause content[11]. Using this structured information, we construct a training dataset consisting of question-answer pairs based on these power grid dispatch regulations clauses to fine-tune an LLM[12]. The next step involves leveraging RAG to introduce more power grid dispatch regulation information, ensuring that it is able to access relevant clauses from the power grid

dispatch regulation documents efficiently[13]. To address the challenge of irrelevant or noisy retrieved documents, we introduce a robust fine-tuning method based on LoRA (Low-Rank Adaptation) [14]. This approach allows the model to effectively distinguish between relevant and irrelevant information, thereby improving its performance in complex scenarios where distractor documents are likely. Figure 1 shows the workflow of the method proposed in this paper.

Our key findings demonstrate that our fine-tuning strategy significantly enhances the performance and robustness of LLMs in power grid dispatch domain, making them more reliable in real-world applications where data retrieval can be imperfect. This approach has broader implications for any domain that requires precise and context-sensitive knowledge extraction, offering a pathway to improving the integration of LLMs in complex, real-world systems[15].

II. METHODOLOGY

In the offline process, the power grid dispatch regulation documents are parsed into structured clause data through data preprocessing. Subsequently, the BGE-M3 model is used to encode all clauses into embedding vectors, which are stored in a vector database. During online querying, the user's query is used to retrieve the top-k most relevant clauses from the embedding database. The query, along with the retrieved clauses, is then used to construct prompt, which are input into the fine-tuned large language model to generate answers. This section first introduces the data preprocessing approach and the method for generating fine-tuning samples, followed by the retrieval-augmented approach, supervised fine-tuning, and finally our robust retrieval augmented generation fine-tuning.

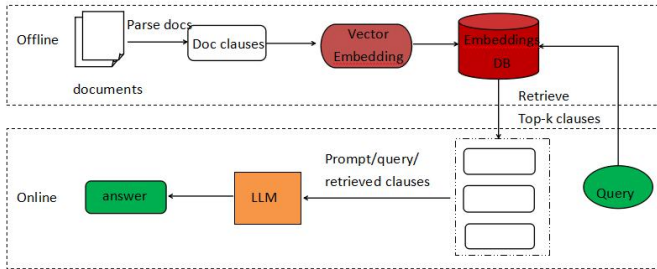


Fig.1 Overall architecture of power grid dispatch regulations question answering

A. DataSets

We collect power grid dispatch regulations documents, which encompass various regulations and requirements within the power dispatch domain. These documents are then processed using PDF parsing tools to extract structured information, including key elements such as titles, clause numbers, and clause contents. Based on the document's title structure, we decompose the dispatch regulations into a series of individual clauses.

For each decomposed dispatch regulation clause, a dataset for fine-tuning the question-answering model is generated using the Qwen2-72B-Instruct model[16]. We design a question and answer pair generation task using prompt engineering, where five relevant questions are formulated for each regulation clause, along with their corresponding answers,

thus resulting in a fine-tuning training dataset, N denotes the total number of clauses, $query_i$ refers to the question generated based on the $clauses_i$, and $answer_i$ represents the corresponding generated answer.

$$T = \{t_i \mid t_i = \{query_i, answer_i, clauses_i\}, i \in N\}$$

After generating the candidate training data, a validation module is employed for post-processing. The validation process filters the dataset based on two primary criteria: the alignment between the question and the reference clause, and the ability of the answer to adequately address the question. Using prompt engineering, we design a filtering task to remove low-quality samples based on the Qwen2-72B-Instruct model, selecting a dataset that meets the criteria as the final fine-tuning training data. Figure 2 illustrates the workflow for automatically generating the fine-tuning dataset.

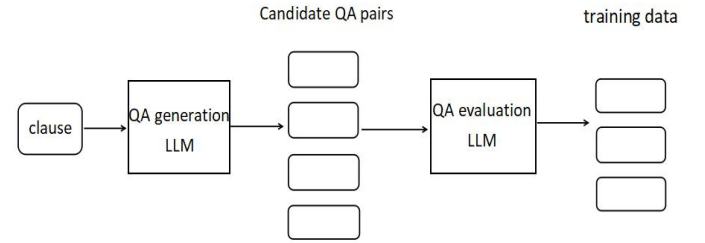


Fig.2 Workflow of fine-tuning dataset auto generation

B. Retrieval Augmented Generation

In this study, we employ the m3e-base model from BGE[17] to convert each clause into a vector representation. Subsequently, the FAISS library[18] is utilized to construct an efficient vector index. Employ an inverted index to cluster similar vectors, grouping semantically similar clauses together. During the retrieval process, inter-group search is performed first, which roughly identifies the group containing the relevant information, thus reducing the search space and improving retrieval efficiency. Once the group containing the answer is identified, a precise match is conducted within the group to ensure the most relevant clauses are retrieved. This approach allows the system to quickly identify the top-k most relevant clauses to the given query, returning these clauses as reference material.

$$Topk(clauses) = \arg \max_k \text{similarity}(query, clause)$$

$$\text{similarity}(query, clause) = \frac{query \cdot clause}{\|query\| \|clause\|}$$

Furthermore, based on the retrieved top-k relevant clauses and the user query, a prompt template is constructed and input into the Baichuan-13B model[19] to generate the final answer. The large language model, leveraging the knowledge acquired through the retrieval module, synthesizes the relevant information and performs reasoning to derive the final response.

C. Supervised Fine-Tuning

Supervised Fine-Tuning (SFT) is a method used to adapt pre-trained models to specific downstream tasks by further training them on labeled datasets. It leverages task-specific

annotations to refine the model's parameters, enabling it to achieve better performance on targeted applications such as text classification, question answering, or generation.

There are various methods for instruction fine-tuning, including full fine-tuning, Prefix-Tuning[20], Prompt-Tuning[21], P-Tuning[22], P-Tuning v2[23], LoRA (low-rank adaptation), QLoRA[24], and others[25]. Considering both effectiveness and efficiency, LoRA is used to fine-tune the Baichuan-13B model, optimizing it by adding low-rank matrices to specific layers of the pre-trained model, rather than updating the entire model. This technique enables rapid adaptation to specific tasks while keeping the large model parameters fixed[26], improving fine-tuning efficiency, reducing computational resource consumption, and effectively preventing overfitting[27].

LoRA assumes that the weight changes during fine-tuning can be approximated by low-rank matrices. Specifically, for a weight matrix W of a layer in the pre-trained model, represent the update as:

$$W' = W + \Delta W$$

Here, ΔW is approximated by the product of two low-rank matrices A and B , A and B are trainable parameter matrices, and their ranks are much smaller than the dimensions of the original weight matrix W .

$$\Delta W = A \cdot B$$

D. Robust Retrieval Augmented Generation Fine-Tuning

To enhance the robustness of the question-answering model, particularly in complex scenarios involving irrelevant information, a robust fine-tuning method is introduced. This method leverages the Low-Rank Adaptation (LoRA) fine-tuning technique and we have meticulously designed it on the training data to enhance the model's ability to recognize and handle interfering information.

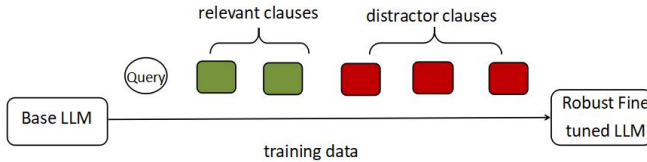


Fig.3 Diagram of Robust Fine-Tuning

During the training process, distractor clauses are introduced to improve the model's robustness in the presence of irrelevant data. For each user query and the corresponding set of retrieved clauses, the model must not only learn to answer the question but also ignore those clauses that are irrelevant to the query. The inclusion of distractor clause aids in optimizing the model's performance, particularly in handling situations with extraneous or misleading information.

$$T' = \{t_i \mid t_i = \{query_i, answer_i, clauses_i, clauses'_i\}, i \in N\}$$

The distractor clauses are randomly selected from clauses other than the relevant ones. A dataset T' is constructed, where each training sample consists of a query, represented by the variable $query_i$, relevant clauses, represented by the variable $clauses_i$, distractor clauses, represented by the variable

$clauses'_i$ and corresponding gold answers, represented by the variable $answer_i$. In this dataset, the order of relevant and distractor clauses is randomized. Then fine-tune the model using LoRA based on this training data. This method effectively simulates the complex retrieval scenarios encountered in practical applications by combining relevant clauses and irrelevant clauses.

III. EXPERIMENT

A. DataSets

The power grid dispatch regulations documents were deconstructed into structured clauses, resulting in about 4,000 clauses. These clauses were encoded into vector representations, forming a clause vector database. For the training dataset, fine-tuning data was generated using the Qwen2-72B-Instruct model, resulting in a total of 10,000 data entries. The test set consisted of 126 questions and gold answers, which were summarized and provided by Power grid dispatch operators.

B. Evaluation Metrics

Accuracy was employed as the effectiveness metric, where the correctness of model-generated responses was manually assessed. The accuracy is calculated as follows:

$$Accuracy = \frac{\#Correct\ Answers}{\#Evaluation\ Set} \times 100\%$$

C. Experimental Setup

The experiments were conducted using the open-source Baichuan2-13B-Chat model. Instruction fine-tuning was performed with the DeepSpeed framework [28] on four Nvidia A6000 GPUs, each with 40 GB memory. We use lora fine-tuning approach with a learning rate of 2E-5, lora_r=16.

D. Results

1) Question-Answer Evaluation

Experimental results show that Baichuan-13B model alone exhibits subpar performance, achieving only 23.8 in Accuracy. This highlights the limitations of general-purpose large language models when applied to power grid dispatch regulations tasks. This indicates that general-purpose large language models are insufficiently trained in the field of power grid dispatch regulations, making it difficult for them to fully comprehend and accurately handle knowledge-based question-answering tasks related to power grid dispatch regulations.

However, upon integrating the RAG module, the model's performance improved significantly, achieving an accuracy of 88.9. This demonstrates that the RAG module effectively enhances the model's response capabilities by incorporating relevant knowledge from retrieved clauses. Further fine-tuning showed minimal improvement in accuracy. Notably, when combined with our robust fine-tuning approach, the model's accuracy increased to 91.2, demonstrating the effectiveness of our method in improving the model's robustness against interference. Compared to traditional fine-tuning, the robust fine-tuning approach enabled the model to better filter

irrelevant information, thus improving both the overall accuracy and reliability of the question-answering system.

TABLE I. QUESTION-ANSWER EVALUATION RESULT

Method	Accuracy
Baichuan-13B	23.8
Baichuan-13B + RAG	88.9
Baichuan-13B+ RAG+ Finetune	89.7
Baichuan-13B+ RAG+ Robust Finetune	91.2

2) Automated Fine-tuning Dataset Generation Evaluation

A random sample of 100 instances from the automated fine-tuning dataset was evaluated based on accuracy. The evaluation criteria included the alignment between the question and the reference clause, as well as the ability of the answer to adequately address the question. The results show that the use of a few-shot learning model for fine-tuning data generation led to a slight improvement in correctness by 2 percentage points, enhancing the model's ability to follow instructions. When a validation model was applied, the correctness of the generated samples increased by 11 percentage points, effectively filtering out samples where the reference clauses or answers did not align with the questions, thereby significantly improving sample quality.

TABLE II. DATASET GENERATION EVALUATION RESULT

Method	Accuracy
Generation model	82
Generation model+few-shot	84
Generation model+few-shot+ validation model	95

IV. CONCLUSION

This paper proposes an enhanced question answering model for power grid dispatch regulations. Firstly, we introduce an automated fine-tuning datasets construction method, which includes document parsing and structured extraction to automatically build high-quality fine-tuning datasets. The retrieval enhancement method implemented through BGE significantly improves the accuracy of the question answering system. Furthermore, by adopting a robust fine-tuning approach for retrieved augmented generation optimized with LoRA, the model's ability to handle complex scenarios involving irrelevant information is strengthened. Experiment results demonstrate that this method enhances the reliability and extensibility of the question answering model. By training the model to ignore distractor information, its performance in real-world environments of power grid dispatching question and answering is improved.

ACKNOWLEDGMENT

This work is supported by the Science and Technology Program of State Grid Corporation of China under Grant No. 5108-202455044A-1-1-ZN.

REFERENCES

- [1] G. Eason, B. Noble, and I. N. Sneddon, "On certain integrals of Lipschitz-Hankel type involving products of Bessel functions," *Phil. Trans. Roy. Soc. London*, vol. A247, pp. 529-551, April 1955. (references)
- [2] Allam A M N, Haggag M H. The question answering systems: A survey[J]. *International Journal of Research and Reviews in Information Sciences (IJRRIS)*, 2012, 2(3).
- [3] Tang Y Y, Lee S W, Suen C Y. Automatic document processing: a survey[J]. *Pattern recognition*, 1996, 29(12): 1931-1952.
- [4] Ghorab M R, Zhou D, O'connor A, et al. Personalised information retrieval: survey and classification[J]. *User Modeling and User-Adapted Interaction*, 2013, 23: 381-443.
- [5] Lewis P, Perez E, Piktus A, et al. Retrieval-augmented generation for knowledge-intensive nlp tasks[J]. *Advances in Neural Information Processing Systems*, 2020, 33: 9459-9474.
- [6] Wu S, Xiong Y, Cui Y, et al. Retrieval-augmented generation for natural language processing: A survey[J]. *arXiv preprint arXiv:2407.13193*, 2024.
- [7] Shi F, Chen X, Misra K, et al. Large language models can be easily distracted by irrelevant context[C]//*International Conference on Machine Learning*. PMLR, 2023: 31210-31227.
- [8] Susnjak T, Hwang P, Reyes N H, et al. Automating research synthesis with domain-specific large language model fine-tuning[J]. *ACM Transactions on Knowledge Discovery from Data*, 2024.
- [9] Ahmadi A. Unravelling the mysteries of hallucination in large Language models: Strategies for Precision in Artificial Intelligence Language Generation[J]. *Asian Journal of Computer Science and Technology*, 2024, 13(1): 1-10.
- [10] Zhang T, Patil S G, Jain N, et al. Raft: Adapting language model to domain specific rag[J]. *arXiv preprint arXiv:2403.10131*, 2024.
- [11] Liu R, Fu R, Xu K, et al. A review of knowledge graph-based reasoning technology in the operation of power systems[J]. *Applied Sciences*, 2023, 13(7): 4357.
- [12] Wu Z, Zhang L, Cao C, et al. Exploring the trade-offs: Unified large language models vs local fine-tuned models for highly-specific radiology nli task[J]. *arXiv preprint arXiv:2304.09138*, 2023.
- [13] Fan S, Guo J, Ma S, et al. Framework and key technologies of human-machine hybrid-augmented intelligence system for large-scale power grid dispatching and control[J]. *CSEE Journal of Power and Energy Systems*, 2023.
- [14] Hu E J, Shen Y, Wallis P, et al. Lora: Low-rank adaptation of large language models[J]. *arXiv preprint arXiv:2106.09685*, 2021.
- [15] Roy P. Enhancing Real-World Robustness in AI: Challenges and Solutions[J]. *Journal of Recent Trends in Computer Science and Engineering (JRTCSE)*, 2024, 12(1): 34-49.
- [16] Chu Y, Xu J, Yang Q, et al. Qwen2-audio technical report[J]. *arXiv preprint arXiv:2407.10759*, 2024.
- [17] Chen J, Xiao S, Zhang P, et al. Bge m3-embedding: Multi-lingual, multi-functionality, multi-granularity text embeddings through self-knowledge distillation[J]. *arXiv preprint arXiv:2402.03216*, 2024.
- [18] Douze M, Guzhva A, Deng C, et al. The faiss library[J]. *arXiv preprint arXiv:2401.08281*, 2024.
- [19] Yang A, Xiao B, Wang B, et al. Baichuan 2: Open large-scale language models[J]. *arXiv preprint arXiv:2309.10305*, 2023.
- [20] LI X L, LIANG P. Prefix-Tuning: optimizing continuous prompts for generation[C]//*Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International*

- Joint Conference on Natural Language Processing. Stroudsburg: ACL, 2021: 4582-4597.
- [21] LESTER B, AL-ROUF R, CONSTANT N. The power of scale for parameter-efficient prompt tuning[C]//Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing. Stroudsburg: ACL, 2021: 3045-3059.
 - [22] LIU X, ZHENG Y, DU Z, et al. GPT understands, too [EB/OL]. [2024-04-03]. <https://arxiv.org/abs/2103.10385>.
 - [23] DING N, QIN Y, YANG G, et al. Parameter-efficient fine tuning of large-scale pre-trained language models[J]. Nature Machine Intelligence, 2023, 5(3): 220-235.
 - [24] DETTMERS T, PAGNONI A, HOLTZMAN A, et al. QLoRA: efficient finetuning of quantized LLMs[C]//Advances in Neural Information Processing Systems 36, New Orleans, Dec 10-16, 2023.
 - [25] Zhang D, Feng T, Xue L, et al. Parameter-Efficient Fine-Tuning for Foundation Models[J]. 2025.
 - [26] LIU Y, LAPATA M. Text summarization with pretrained encoders[C]//Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing. Stroudsburg: ACL, 2019: 3730-3740.
 - [27] VASWANI A, SHAZEER N, PARMAR N, et al. Attention is all you need[C]//Advances in Neural Information Processing Systems 30, Long Beach, Dec 4-9, 2017: 5998-6008.
 - [28] Aminabadi R Y, Rajbhandari S, Awan A A, et al. DeepSpeed-inference: enabling efficient inference of transformer models at unprecedented scale[C]//SC22: International Conference for High Performance Computing, Networking, Storage and Analysis. IEEE, 2022: 1-15.